MILITARY SPECIFICATION

RESCUE HAND TOOL, PNEUMATIC

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for one type of pneumatic rescue hand tool.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
UU- P- 268	Paper, Kraft, Untreated, wrapping
PPP-B-576	Boxes, Wood, Cleated, Veneer, Paper overlaid
PPP-B-601	Boxes, Wood, Cleated- Plywood
PPP-B-636	Box, Fiberboard
PPP-B-665	Box, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676	Boxes, Setup

Military

MIL-W-4088	Webbing, Textile, Woven Nylon

FSC 4240

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Military (Continued)

MIL-W-5044	Walkway Compound, Nonslip and Walkway Matting, Nonslip
MIL-P-5425	Plastic Sheet, Acrylic, Heat Resistant
MIL-P-8184	Plastic Sheet, Acrylic, Modified
MIL-C-81575	Cylinder, Nitrogen, for Rescue Hand Tool

STANDARDS

Federal

FED-STD-151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Air Systems Command

64 A83C41	Parts Index, Rescue Hand Tool Assembly
64A83H18	Holster Assembly, Rescue Hand Tool

(When requesting any of the applicable documents, refer to both title and number. .411 requests should be made via the cognizant Government quality assurance representative. Copies of this specification and other unclassified specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The rescue hand tools furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids. (See 4.3 and 6. 3).

3.2 First article - The rescue hand tools furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

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3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings.

3.3.1 Metal parts - All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.3.2 Protective treatment - Materials used in the construction of the rescue hand tools that are subject to deterioration when exposed to the climatic and environmental conditions that will occur during service use, shall be protected against such deterioration in such a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip or scale with age or extremes of climate and environmental conditions shall be avoided.

3.4 Construction - The rescue hand tools shall be constructed in accordance with the applicable drawings specified on Naval Air Systems Command Drawing No. 64A83C41.

3.5 Holster - Each rescue hand tool shall be furnished with a holster conforming to NASC Drawing No. 64A83H18.

3.6 Nitrogen cylinder - The nitrogenn cylinder used to operate the rescue hand tool shall conform to MIL-C-81575.

3.7 Performance -

3.7.1 Proof pressure - The rescue hand tool, when tested as specified in 4.6.2, shall not show any signs of leakage.

3. 7.2 Leakage - The rescue hand tool, when tested as specified in 4.6.3, shall not show signs of leakage after being submerged for a period of 5 minutes.

3. ’7.3 Operation - The rescue hand tool, when tested as specified in 4.6.4, shall be capable of making a rninimum of four cuts through double thicknesses of webbing conforming to MIL-W-4088, Type XIII and one cut through a 7/32 inch diameter 19 x 7 stainless steel cable.

3. 7,4 Impact - The rescue hand tool, when tested as specified in 4.6.5, shall not show signs of material failure, leakage or malfunction.

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3.8 Finish -

3.8.1 Surface treatment - Aluminum parts shall be cleaned and anodized as specified on the applicable drawing.

3.8.2 Painting - Walkway coating conforming to MIL-W-5044 shall be applied to the handle area, as indicated on the applicable drawing, to provide a nonslip surface.

3.9 Workrnanship - The rescue hand tools shall be uniform in quality and shall be free from defects, irregularities or foreign matter which could adversely affect safety, perfomance, durability or reliability.

4. QUALITY ASSURANCE PROVISIONS

4,1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the rescue hand tools shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of tests and examinations performed on samples submitted for approval as qualified products (see 4. 3).

(b) First article inspection - First article inspection consists of examinations and tests performed on rescue hand tools which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4. 4).

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with requirements set forth in this specification (see 4. 5).

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4.3 Qualification inspection - Qualification inspection shall consist of all tests and examinations listed in this specification.

4.3.1 Qualification inspection samples - The qualification inspection samples shall consist of two rescue hand tools. The qualification samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for Qualification Inspection
RESCUE HAND TOOL, PNEUMATIC
Manufacturer's designation or number
Name of Manufacturer
Submitted by (name) (date) for qualification inspection in accordance with the requirements of Specification MIL-R-81574 under authorization (reference authorizing letter).

4.4 First article inspection - The first article inspection shall consist of the following examinations and tests:

Visual examination
Dimensional examination
Proof pressure
Leakage
Operation
Impact

4.4.1 First article samples - Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacture. r shall submit two rescue hand tools. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. when a manufacture r is in continuous production of these rescue hand tools from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the procuring activity (see 6. 2). Approval of the first article inspection sample or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished the Government as directed by the contracting officer (see 6. 2).

4.4.2 Upon completion of the first article inspection, all the applicable inspection reports, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the cognizant

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Government Activity. One rescue hand tool shall be returned to the manufacturer for use in monitoring production. The other rescue hand tool will be consumed or destroy cd in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection : The quality conformance inspection shall consist of the following:

Visual examination
Dimensional examination
Proof pressure
Leakage
Operation
Impact
Preparation for delivery

4. 5.1 Sampling -

4.5. 1.1 Inspection lot -

4.5. 1.1.1 Rescue hand tools - An inspection lot shall be expressed in units of one rescue hand tool, made essentially under the same conditions and from the same materials and components. The sample unit shall be one rescue hand tool.

4.5. 1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing rescue hand tools fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing rescue hand tools fully prepared for delivery with the exception that it need not be sealed.

4. 5.1.2 Sampling for tests and examinations of rescue hand tools - The sample size, acceptance criteria, tests, and examinations required for the rescue hand tools shall be as specified in Table I.

4.6 Inspection methods -

4.6.1 Visual examination -

4.6. 1.1 Rescue hand tools - Each rescue hand tool selected in accordante with Table I shall be examined visually to determine conformance to this specification. The classification of defects, Table II, shall be used to classify the defects found. For dimensional defects, each rescue hand tool selected as a sample unit from the lot shall be thoroughly checked dimensionally to determine conformance to this specification. The classification of defects, Table III, shall be used to classify the defects found.

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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, EXAMINATIONS

INSPECTION	METHOD PARAGRAPH	SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
Visual examination	4.6.1.1	Every unit for critical defects. Inspection Level II for minor defects	Reject all units with any critical defects and an Acceptable Quality Level of 2.5 defects per 100 units for minor defects
Dimensional examination	4.6.1.1	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Proof pressure	4.6.2	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Leakage	4.6.3	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Operation	4.6.4	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Impact	4.6.5	Inspection Level S-2	An Acceptable Quality Level of 2.5 defects per 100 units
Preparation for delivery	4.6.1.2	Inspection Level S-2	An Acceptable Quality Level of 4.0 defects per 100 units

1/ The sampling plan acceptance number shall apply collectively to all the characteristics within a stated acceptable quality level.

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TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE TOOL

DEFECTS	MINOR	CRITICAL
Incorrect configuration		X
Incorrect or defective threads		X
Incorrect, incomplete, or faulty finish	X	
Any functioning part that works with difficulty		X
Flaw in the material		X
Incorrect or incomplete marking	X	
Loose particles inside the rescue hand tool		X
Rust or scale		X
Existence of any foreign material, solid or liquid		X

TABLE III

LIST OF DEFECTS FOR FINISHED DIMENSIONS

EXAMINE	DEFECT
Measure all visible components of the pneumatic rescue hand tool assembly	Any measurement deviating from the dimensions and tolerances as specified in this specification and applicable ACED drawings shall be classified as a dimensional defect.

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TABLE IV

CLASSIFICATION OF DEFECTS FOR PREPARATION FOR DE LIVERY

ITEM	DEFECT
Contents	More than one complete assembly in the same container. All components not of the same size.
Exterior and interior	Missing, incomplete, incorrect, illegible; of improper size, location, sequence, or method of application; exterior and interior markings different.
Packaging and packing materials	inadequate application of the components such as incomplete closure of unit package, intermediate package, case liners, container flaps, loose strappings, etc. ; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirement.

4.6. 1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing rescue hand tools, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

4.6.2 Proof pressure test - The gas storage area of the rescue hand tool shall be subjected to an internal pressure of 4000 psig for one minute. The rescue hand tool shall pass the requirements specified in 3.7.1.

4.6.3 Leakage - The complete rescue hand tool shall be charged using a nitrogen cylinder; then, without depressing the trigger, completely submerged in clear water to a depth of one inch The rescue hand tool shall pass the requirements specified in 3.7.2. With the rescue hand tool still submerged, two thicknesses of webbing conforming to MIL-W-4088, Type XIII shall be placed in the opening, and the trigger squeezed and held. The rescue hand tool shall pass the requirements specified in 3.7.2.

4.6.4 Operation - With a test blade installed, the rescue hand tool shall be pressurized using a nitrogen cylinder. The rescue hand tool shall be submerged in one foot of water and operated. The rescue hand tool shall meet the requirements specified in 3.7.3.

4.6.5 Impact test - The rescue hand tool shall be used as a hammer to break through a piece of acrylic plastic sheet, MIL-P-5425 finish A, or modified acrylic plastic sheet, MIL-P-8184 finish A, 0.312 thick. The acrylic

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plastic specimen shall be 4 feet x 4 feet and shall be clamped securely to a rigid frame. The aft face of the hook assembly in the area directly behind the blade opening shall be used as the hammer head. The rescue hand tool shall be held in the normal gripping area with one hand. Upon completion of the impact test, the rescue hand tool shall be checked in accordance with 4.6.3. The rescue hand tool shall meet the requirements specified in 3. 7.2 and 3.7.4.

5, PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C, as specified (see 6. 2). Other than as specified in paragraph 5.1.1.1, no preservation is required.

5. 1.1 Level A -

5.1.1.1 Unit package - Each rescue hand tool shall be unit packaged with a suitable Kraft paper wrap, UU-P-268, and placed within a setup or fiberboard box conforming to PPP-B-636, PPP-B-665, or PPP-B-676 Type I or II, Class I, maintaining a snug even fit.

5. 1.1.2 Intermediate package - Intermediate packages shall contain the number of unit packages specified by the procuring activity. The intermediate container shall be flberboard boxes conformlng to PPP-B-636.

5. 1.2 Level C -

5.1.2.1 Preservation and packaging - Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing - Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A - Rescue hand tools packaged as specified in 5. 1.1 shall be packed in overseas type cleated fiberboard, cleated plywood or wood cleated, fiberboard, paper-overlaid containers conforming to PPP-B-576, PPP-B-591, PPP-B-601, or PPP-B-636. The containers shall be closed and strapped in accordance with the requirements of the appendix of the applicable box specification. The gross weight of the exterior shipping container when fully packed shall not exceed approximately 200 pounds. Plywood, when used, shall conform to NN-P-515, Type I or II, Class 2.

5.2.2 Level B - Rescue hand tools packaged as described in 5.1.1 shall be packed in domestic type containers conforming to PPP-B-576, PPP-B-591, PPP-B-601, or PPP-B-636. The gross weight of wood or wood cleated containers shall not exceed 200 pounds.

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5.2.3 Level C - Rescue hand tools packaged as specified in 5. 1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking - Unit packages, and intermediate containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The rescue hand tool covered by this specification is intended to be used by helicopter crews, engaged in open sea rescue, to free survivors from shroud lines and any other equipment that interferes with rescue operations.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Quantity desired

(c) Whether first article inspection is waived (see 4.4.1)

(d) Name and address of first article inspection laboratory (see 4.4. 1)

(t?) Selection of applicable levels of packaging and packing (see 5.1, 5.2 and 5.3)

6.3 Qualification - With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacture's are urged to arrange to have the products that they p repose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington D. C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Johnsville, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

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6.3.1 When requested, the manufacturer will submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

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DEPARTMENT OF THE NAVY

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Washington, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-81574(AS) Rescue Hand Tool, Pneumatic

ORGANIZATION *(of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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